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What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 1: Spacetime, Fields and Observables

Group-B

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1 Objective and spacetime convention

Project question. How does light–spectator entanglement depend on mass and mixing, and does more entanglement imply a stronger heavy-field three-point signal? Quadratic mixing can produce Gaussian entanglement; a connected three-point signal also needs a non-Gaussian interaction.

Scope. We fix the background, action, normalisation and observables, then derive and check the specified single-field squeezed reference. Coupled-field entropy and the spectator bispectrum belong to later steps.

1.1 Expansion and horizon crossing

We use natural units $\hbar = c = 1$ and an exact de Sitter calculational background,

$$\mathrm{d}s^2 = a^2(\eta)(-\mathrm{d}\eta^2 + \mathrm{d}\boldsymbol{x}^2), \qquad a(\eta) = -\frac{1}{H\eta}, \qquad \eta < 0, \quad H > 0. \tag{1}$$

Here H is the constant Hubble expansion rate and a is the scale factor. In this de Sitter patch, conformal time runs from $-\infty$ to 0, while cosmic time runs from $-\infty$ to $+\infty$, with $\mathrm{d}t = a\,\mathrm{d}\eta$. A prime denotes differentiation with respect to η. As η increases towards zero, a grows.

For a fixed comoving wavenumber $k = |\boldsymbol{k}| > 0$, define

$$\boxed{x = -k\eta = \frac{k}{aH}.} \tag{2}$$

The physical wavelength is $\lambda_{\mathrm{ph}} = a\lambda_{\mathrm{com}} = 2\pi a/k$, so the physical wavenumber is $k_{\mathrm{ph}} = k/a$. The Hubble radius is $R_H = H^{-1}$ in our units. Thus $x = k_{\mathrm{ph}}/H = k_{\mathrm{ph}}R_H = 2\pi R_H/\lambda_{\mathrm{ph}}$ compares the physical wavenumber with the expansion rate. Evolution proceeds from large to small x. Numerically we set $H = 1$, retaining m/H and ρ/H.

Regime	Condition	Interpretation
Inside the horizon	$x \gg 1$	Reduced wavelength $a/k \ll R_H$.
Horizon crossing	$x = 1$	$k = aH$, or $\lambda_{\mathrm{ph}} = 2\pi R_H$.
Outside the horizon	$x \ll 1$	Reduced wavelength $a/k \gg R_H$.

Here the usual horizon convention compares a/k with H^{-1}; retaining a literal wavelength $2\pi a/k$ introduces the corresponding factor of 2π.

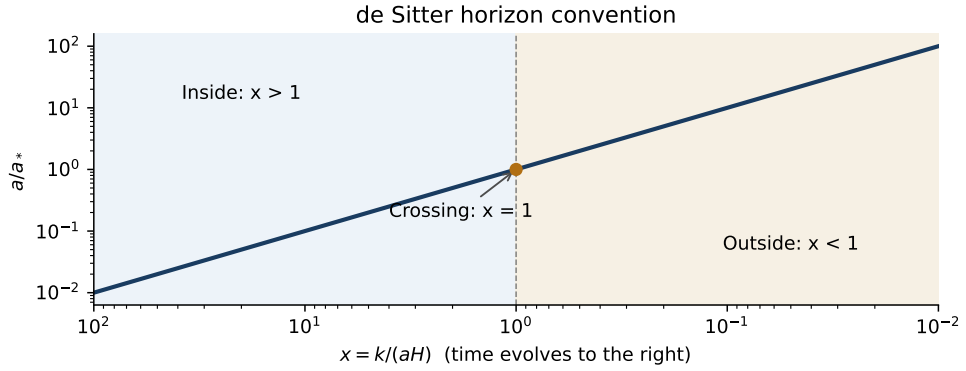


Figure 1: For $H = k = 1$, $a/a_* = 1/x$, where $a_* = k/H$. The axis follows forward time. This checks the background convention, not field-mode evolution.

2 Fields, action and physical parameters

We use the quadratic action

$$S_2 = \int \mathrm{d}\eta\, \mathrm{d}^3x \left\{ \frac{a^2}{2} \left[(q')^2 - (\boldsymbol{\nabla} q)^2 + (\sigma')^2 - (\boldsymbol{\nabla}\sigma)^2 \right] - \frac{a^4 m^2}{2}\sigma^2 + a^3 \rho\, q'\sigma \right\}. \tag{3}$$

Symbol	Meaning
q	Light scalar perturbation, proportional to the curvature perturbation ζ at linear order.
σ	Spectator scalar field; it is the additional degree of freedom to be traced out when studying the light subsystem.
m	Spectator mass parameter in the quadratic action.
ρ	Constant derivative-mixing strength multiplying $q'\sigma$.
$M = m/H$	Dimensionless mass, used when comparing masses with the expansion rate.
$g = \rho/H$	Dimensionless mixing strength.

2.1 Reading the action

The terms containing time derivatives are kinetic terms. The gradient terms describe spatial variation, and the $m^2\sigma^2$ term supplies the spectator mass. The last term couples the time variation of q to σ. Thus, at nonzero ρ, the two fields do not evolve independently. At $\rho = 0$ the quadratic action separates into a massless light field and a massive spectator.

The mixing contains a derivative. Consequently, the conjugate momentum is $\pi_q = \partial\mathcal{L}/\partial q' = a^2 q' + a^3\rho\sigma$: the mixing supplies an additional term. The full Legendre transform and canonical evolution are the subject of Step 3; here we retain the action without expanding or dropping any terms.

2.2 Dimensions and normalisation

With dimensionless a and coordinates of inverse-mass dimension, the action is dimensionless when

$$[q] = [\sigma] = [H] = [m] = [\rho] = \text{mass}, \qquad [M] = [g] = 1. \tag{4}$$

For example, $[q'] = \text{mass}^2$ and $[\rho q'\sigma] = \text{mass}^4$, matching the kinetic and mass terms. The integration measure has dimension mass^{-4}.

Write the linear curvature conversion as $q = C\zeta$, where ζ is dimensionless and C has dimension mass. No numerical value of C is fixed for this project. Power spectra and bispectra acquire different powers of C, so we keep q and ζ observables distinct. A specific inflationary model would require its background normalisation; the conversion factors are given in Section 3.2.

Project assumptions. We use the specified two-field model on de Sitter space without the full gravitational cubic interactions. Entanglement entropy is calculated in the quadratic theory, which retains terms of total degree two in the fields and their derivatives. The spectator-induced bispectrum is calculated separately, at leading order in a specified cubic coupling.

3 Fourier convention and observables

We use the following Fourier convention for both fields:

$$q(\eta, \boldsymbol{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, q_{\boldsymbol{k}}(\eta) e^{i\boldsymbol{k}\cdot\boldsymbol{x}}, \qquad [a_{\boldsymbol{k}}, a^\dagger_{\boldsymbol{p}}] = (2\pi)^3\delta^3(\boldsymbol{k}-\boldsymbol{p}). \tag{5}$$

For a real field, $q_{-\boldsymbol{k}} = q^\dagger_{\boldsymbol{k}}$: opposite Fourier modes are related by a reality condition. A generic nonzero pair is described by independent real cosine and sine sectors. Step 2 fixes their counting when calculating entropy.

Translation invariance gives the momentum-conserving delta functions:

$$\langle q_{\boldsymbol{k}} q_{\boldsymbol{p}}\rangle = (2\pi)^3\delta^3(\boldsymbol{k}+\boldsymbol{p})P_q(k), \tag{6}$$

$$\langle q_{\boldsymbol{k}_1} q_{\boldsymbol{k}_2} q_{\boldsymbol{k}_3}\rangle_c = (2\pi)^3\delta^3(\boldsymbol{k}_1+\boldsymbol{k}_2+\boldsymbol{k}_3)B_q(k_1,k_2,k_3). \tag{7}$$

P_q measures the strength of two-point correlations at a given wavenumber. B_q measures the connected correlation of three Fourier modes; the subscript c means that products of lower-order correlations have been subtracted. Momentum conservation requires the three vectors to close into a triangle. A squeezed triangle has $k_L \ll k_S$, equivalently $\lambda_L \gg \lambda_S$, with two nearly equal short-wavelength sides.

3.1 Power spectrum versus dimensionless power

For the curvature perturbation, define

$$\Delta^2_\zeta(k) = \frac{k^3}{2\pi^2}P_\zeta(k), \qquad n_s - 1 = \frac{\mathrm{d}\ln\Delta^2_\zeta}{\mathrm{d}\ln k}. \tag{8}$$

For constant n_s, $\Delta^2_\zeta \propto k^{n_s-1}$, so

$$P_\zeta(k) \propto k^{n_s-4}. \tag{9}$$

The exponent of P_ζ is therefore $n_s - 4$, rather than $n_s - 1$. Mixing these two spectra would give the wrong squeezed-limit coefficient.

3.2 Keeping the field conversion explicit

At a common evaluation time, a scale-independent conversion $q = C\zeta$ gives

$$P_q = C^2 P_\zeta, \qquad B_q = C^3 B_\zeta, \qquad \frac{B_\zeta}{P_\zeta(k_L)P_\zeta(k_S)} = C\,\frac{B_q}{P_q(k_L)P_q(k_S)}. \tag{10}$$

In this Fourier convention, $[P_q] = \text{mass}^{-1}$ and $[B_q] = \text{mass}^{-3}$, whereas $[P_\zeta] = \text{mass}^{-3}$ and $[B_\zeta] = \text{mass}^{-6}$. A useful later dimensionless spectator quantity is $HB_q/(P_qP_q)$, for the later spectator calculation. It equals the curvature ratio only if the conversion factor is $C = H$.

Gaussian baseline. In a centred Gaussian state, $\langle q\rangle = 0$ and all odd moments vanish by Wick's theorem, including the connected three-point function. Quadratic evolution preserves Gaussianity, while mixing can still generate light–spectator entanglement. The value derived next comes from a separate single-clock curvature reference, rather than from Eq. (3) alone.

4 Derivation of the single-field squeezed reference

In the single-clock attractor setting, a sufficiently long curvature mode is approximately constant across a short-scale region. Its leading effect is a spatial dilation, which rescales the short-mode wavenumber to $ke^{-\zeta_L}$. The short-scale spectrum conditioned on a small long mode ζ_L transforms as [1, 2]

$$P_\zeta(k \mid \zeta_L) = e^{-3\zeta_L} P_\zeta(ke^{-\zeta_L}). \tag{11}$$

The argument changes with the local spatial scale; the prefactor $e^{-3\zeta_L}$ accounts for the three-dimensional Fourier volume. Expanding both factors to first order gives

$$e^{-3\zeta_L} = 1 - 3\zeta_L + O(\zeta_L^2), \tag{12}$$

$$P_\zeta(ke^{-\zeta_L}) = P_\zeta(k) - \zeta_L k\partial_k P_\zeta(k) + O(\zeta_L^2), \tag{13}$$

$$\delta P_\zeta(k) = -\zeta_L(3 + k\partial_k)P_\zeta(k) + O(\zeta_L^2). \tag{14}$$

The squeezed bispectrum correlates the long mode with the modulation of the short-mode power. Correlating δP_ζ with ζ_L supplies $P_\zeta(k_L)$. With the overall momentum delta function removed,

$$\lim_{k_L/k_S\to 0} B_\zeta(k_L, k_S, k_S) = -P_\zeta(k_L)(3 + k_S\partial_{k_S})P_\zeta(k_S). \tag{15}$$

Now insert $P_\zeta(k) = A_0 k^{n_s-4}$, with A_0 independent of k:

$$k\partial_k P_\zeta(k) = (n_s - 4)P_\zeta(k), \tag{16}$$

$$(3 + k\partial_k)P_\zeta(k) = [3 + n_s - 4]P_\zeta(k) = (n_s - 1)P_\zeta(k). \tag{17}$$

Dividing Eq. (15) by the two power spectra yields the required result:

$$\boxed{\lim_{k_L/k_S\to 0} \frac{B_\zeta(k_L, k_S, k_S)}{P_\zeta(k_L)P_\zeta(k_S)} = -(n_s - 1) = 1 - n_s = 0.035 \quad (n_s = 0.965).} \tag{18}$$

The ratio is positive for the specified red tilt. At $n_s = 1$ it vanishes, providing a useful control. The number 0.035 is this particular dimensionless ratio; it is not the unnormalised bispectrum or a direct estimate of a spectator mass.

4.1 Applicability of the reference

The relation requires a long adiabatic mode that acts as a coordinate rescaling in a single-clock attractor regime. It need not hold in multifield or non-attractor models. Our spectator supplies an additional degree of freedom whose interactions can produce squeezed signals beyond this single-field reference.

The prescribed $n_s = 0.965$ is an external input. An uncoupled massless scalar with the usual vacuum in exact de Sitter space has a scale-invariant late-time spectrum, corresponding to $n_s = 1$. Thus 0.035 is the normalised ratio for the chosen tilted single-field reference, not a contribution generated by our quadratic two-field action. We do not add it to a later spectator bispectrum.

5 Reproducible numerical validation

The accompanying Python script implements two numerical checks of the analytic coefficient. For an arbitrary positive amplitude A and pivot k_*, it uses

$$P_\zeta(k) = \frac{A}{k^3}\left(\frac{k}{k_*}\right)^{n_s-1}. \tag{19}$$

The normalisation is illustrative; no observational amplitude is fitted. The script evaluates 121 logarithmically spaced values with $10^{-3} \le k/k_* \le 10^3$, using $A = k_* = 1$.

Check 1: differentiate the power directly. With fractional step h,

$$R_h(k) = -3 - \frac{P_\zeta(k(1+h)) - P_\zeta(k(1-h))}{2hP_\zeta(k)}. \tag{20}$$

R_h approximates $-3 - k\partial_k P_\zeta/P_\zeta$, testing the consistency coefficient directly. The central difference has an $O(h^2)$ truncation error. Refinement gives:

Fractional step h	$\max_k \lvert R_h - (1-n_s)\rvert$
1.000×10^{-2}	1.028×10^{-3}
5.000×10^{-3}	2.569×10^{-4}
2.500×10^{-3}	6.423×10^{-5}
1.250×10^{-3}	1.606×10^{-5}
1.000×10^{-4}	1.028×10^{-7}
1.000×10^{-5}	1.053×10^{-9}

Check 2: differentiate the finite dilation. Define $P(k \mid \epsilon)$ using Eq. (11) and compute

$$R_\epsilon(k) = \frac{P_\zeta(k \mid \epsilon) - P_\zeta(k \mid -\epsilon)}{2\epsilon P_\zeta(k)}. \tag{21}$$

R_ϵ measures the power spectrum's response to a small long-mode dilation. At $h = 10^{-5}$ and $\epsilon = 10^{-4}$, the maximum absolute errors are 1.053×10^{-9} and 3.886×10^{-12}, respectively. Both are below the configured absolute tolerance 5×10^{-8}. These are numerical differentiation errors in a known benchmark, not uncertainties on a measured cosmological signal.

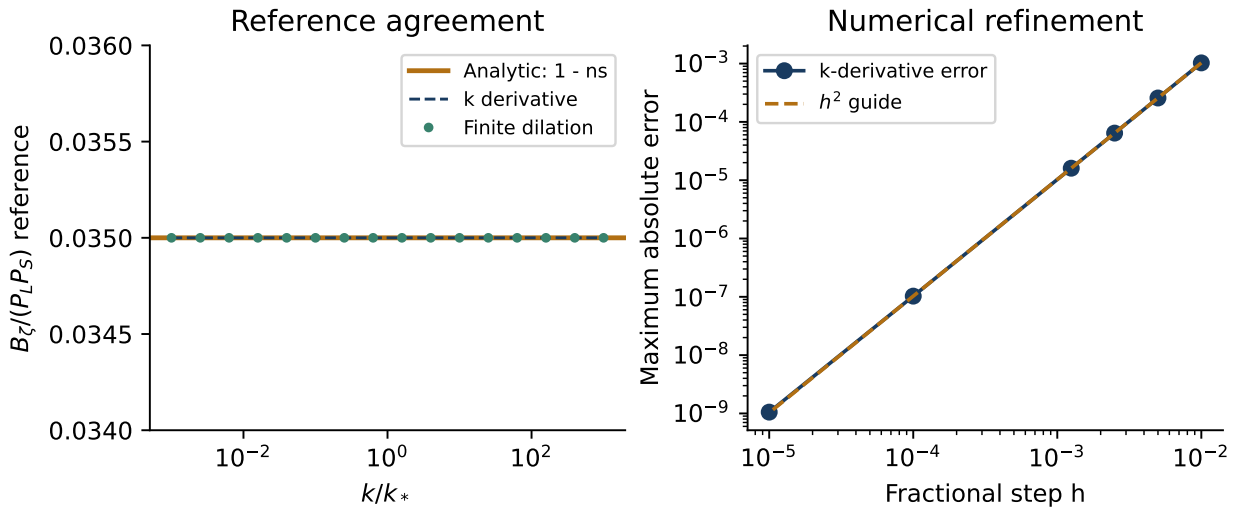


Figure 2: Left: the analytic value 0.035 agrees with both numerical routes over six decades of k. Right: direct differentiation improves approximately as h^2. The calculations check the response identity; they do not evaluate a finite-squeezing or spectator bispectrum.

6 Results, reproducibility and next step

6.1 Validation outcome

All 10 configured checks passed. In addition to the two derivative routes, the script checks the $n_s = 1$ zero reference, red- and blue-tilt controls, cancellation of the arbitrary amplitude and pivot, second-order refinement, and the horizon identity $x = k/(aH)$. These tests verify different features of the implementation, though both derivative routes share the same input power spectrum; they are not two independent physical predictions.

Established in Step 1. The background, Fourier convention, dimensions, action and observable normalisations are fixed. The required squeezed reference is 0.035 at $n_s = 0.965$, with numerical checks supporting the algebra. No entropy map, coupled-field power spectrum, heavy-field frequency or mass estimate is claimed at this stage.

6.2 Code and reproduction

The implementation is organised in a `step1/` folder. From the repository root, run:

```
python3 -m pip install -r step1/requirements.txt
python3 step1/step1.py -config step1/config.json
```

The two supporting data tables and validation summary are included in `step1/results/`, with PNG previews in `step1/figures/`. The README explains each figure and table. Running the script regenerates these outputs and vector PDFs for report preparation. Inputs are in `config.json`; the summary records checks, software versions and code/configuration hashes.

The calculation used Python 3.12.14, NumPy 2.3.5 and Matplotlib 3.10.8. It is deterministic and uses no random sampling.

Code availability. Step 1 code folder.

Python script Configuration Dependencies Run instructions.

Numerical results Figures Validation summary.

Code and outputs: `28ea478`.

6.3 Immediate next step

Step 2 will first validate Gaussian entropy for two ordinary coupled oscillators. It will reproduce the assigned $\nu = 0.502816542$ and $s = 0.0193599304$ benchmark, test zero coupling, increase the coupling to 0.6, and check invariance under a local canonical rescaling. Entropy will be reported in nats per real sector, with the nonzero $\pm\boldsymbol{k}$ pair carrying twice that value. Subsequent steps will derive canonical equations before evolving cosmological modes.

References

[1] J. Maldacena, *Non-Gaussian features of primordial fluctuations in single field inflationary models*, JHEP 05 (2003) 013. arXiv:astro-ph/0210603.

[2] P. Creminelli and M. Zaldarriaga, *Single field consistency relation for the 3-point function*, JCAP 10 (2004) 006. arXiv:astro-ph/0407059.

7 Contributions

Member	Contribution
Priyansh Rawat	Physical setting, quadratic action and shared conventions; review of numerical checks and interpretation.
Samarth Bhatnagar	Fourier conventions, dimensional analysis and the q–ζ conversion; review of the squeezed-limit derivation.
Mayank Sinha	Numerical implementation, validation, data and figures; reproduction instructions and report consistency.